ECE 102 Homework 6

Edi Zhang

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1. Inverse Fourier Transforms and Duality (10 points)

(a) (5 points) Find the inverse Fourier transform of the signal

Solution:

By observation, $\angle X(j\omega) = -\omega/2$. Then

$$X(j\omega) = |X(j\omega)|e^{j\angle X(j\omega)} \tag{1}$$
$$= |X(j\omega)|e^{-j\omega/2} \tag{2}$$

To find the inverse Fourier transform:

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} |X(j\omega)|e^{-j\omega/2}e^{j\omega t}d\omega \tag{3}$$
$$= \frac{1}{2\pi}\left[\int_{-3}^{-2} e^{j\omega(t-\frac{1}{2})}d\omega + \frac{1}{2}\int_{-2}^{2} e^{j\omega(t-\frac{1}{2})}d\omega + \int_{2}^{3} e^{j\omega(t-\frac{1}{2})}d\omega\right] \tag{4}$$
$$= \frac{1}{2\pi}\left[\left[\frac{e^{j\omega(t-\frac{1}{2})}}{j(t-\frac{1}{2})}\right]_{-3}^{-2} + \frac{1}{2}\left[\frac{e^{j\omega(t-\frac{1}{2})}}{j(t-\frac{1}{2})}\right]_{-2}^{2} + \left[\frac{e^{j\omega(t-\frac{1}{2})}}{j(t-\frac{1}{2})}\right]_{2}^{3}\right] \tag{5}$$
$$= \frac{1}{2\pi j(t-\frac{1}{2})}\left[e^{-2j(t-\frac{1}{2})} - e^{-3j(t-\frac{1}{2})} + \frac{1}{2}(e^{2j(t-\frac{1}{2})} - e^{-2j(t-\frac{1}{2})}) + e^{3j(t-\frac{1}{2})} - e^{2j(t-\frac{1}{2})}\right] \tag{6}$$
$$= \frac{1}{\pi(t-\frac{1}{2})}\left(\sin(-2t+1) + \frac{1}{2}\sin(2t-1) + \sin\left(3t - \frac{3}{2}\right)\right) \tag{7}$$
$$= \frac{1}{2\pi(t-\frac{1}{2})}\left(2\sin\left(3\left(t-\frac{1}{2}\right)\right) - \sin\left(2\left(t-\frac{1}{2}\right)\right)\right) \tag{8}$$

(b) (5 points) Find the inverse Fourier transform of (hint: use duality)

$$X(j\omega) = \begin{cases} 1 - |\omega|, & |\omega| \leq 1 \\ 0, & \text{otherwise} \end{cases} \tag{9}$$

Solution:

We have that $\mathcal{F}[x(t)] = X(j\omega)$ for some $x(t)$ that we seek to find. By duality,

$$\mathcal{F}[X(t)] = 2\pi(x(-j\omega)) \tag{10}$$

Then, by the Catalog Fourier Transforms,

$$\mathcal{F}[X(t)] = \mathcal{F}[\Delta(t)] = \text{sinc}^2(\omega/2\pi) \tag{11}$$

$$\implies x(-j\omega) = \frac{1}{2\pi}\text{sinc}^2(\omega/2\pi) \tag{12}$$

Since sinc is an even function,

$$x(t) = \frac{1}{2\pi}\text{sinc}^2(t/2\pi) \tag{13}$$

2. Frequency Response (37 points)

(a) (18 points) Consider the LTI system whose response to unknown input $x(t)$ is:

$$y(t) = (4e^{-t} - 2e^{-2t})u(t) \tag{14}$$

Intermediate signal:

$$y_1(t) = 2e^{-t}u(t) \tag{15}$$

The differential equation:

$$\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 6y(t) = 3x(t) \tag{16}$$

i. Find $H(j\omega)$ of the overall system $h(t)$

Solution:

Recall that if $y(t) = h(t) * x(t)$, then $Y(j\omega) = H(j\omega) \cdot X(j\omega)$. Solving for $Y(j\omega)$:

$$Y(j\omega) = 4\mathcal{F}[e^{-t}u(t)] - 2\mathcal{F}[e^{-2t}u(t)] \tag{17}$$

$$= \frac{4}{1+j\omega} - \frac{2}{2+j\omega} \tag{18}$$

$$= \frac{4(2+j\omega) - 2(1+j\omega)}{(1+j\omega)(2+j\omega)} \tag{19}$$

$$= \frac{6+2(j\omega)}{(1+j\omega)(2+j\omega)} \tag{20}$$

$$= \frac{2(3+j\omega)}{(1+j\omega)(2+j\omega)} \tag{21}$$

Note also that:

$$3X(j\omega) = (j\omega)^2 Y(j\omega) + 5j\omega Y(j\omega) + 6Y(j\omega) \tag{22}$$

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{3}{(j\omega)^2 + 5(j\omega) + 6} \tag{23}$$

$$= \frac{3}{(j\omega+2)(j\omega+3)} \tag{24}$$

Then

$$X(j\omega) = \frac{Y(j\omega)}{H(j\omega)} \tag{25}$$

$$= \frac{2(3+j\omega)}{(1+j\omega)(2+j\omega)} \cdot \frac{(j\omega+2)(j\omega+3)}{3} \tag{26}$$

$$= \frac{2(j\omega+3)^2}{3(j\omega+1)} \tag{27}$$

ii. Find $H_1(j\omega)$ and $H_2(j\omega)$

Solution:

Because these are all LTI systems, $y_1(t) = h_1(t) * x(t)$. Then

$$Y_1(j\omega) = H_1(j\omega) \cdot X(j\omega) \tag{28}$$

$$\frac{2}{1+j\omega} = H_1(j\omega) \cdot \frac{2(j\omega+3)^2}{3(j\omega+1)} \tag{29}$$

$$\implies H_1(j\omega) = \frac{3}{(j\omega+3)^2} \tag{30}$$

$$H(j\omega) = H_2(j\omega) \cdot H_1(j\omega) \tag{31}$$

$$\frac{3}{(j\omega+2)(j\omega+3)} = H_2(j\omega) \cdot \frac{3}{(j\omega+3)^2} \tag{32}$$

$$\implies H_2(j\omega) = \frac{3+j\omega}{2+j\omega} \tag{33}$$

$$\tag{34}$$

iii. Find $h(t)$, $h_1(t)$, $h_2(t)$

Solution:

$$H(j\omega) = \frac{3}{(j\omega+2)(j\omega+3)} = \frac{A}{j\omega+2} + \frac{B}{j\omega+3} \tag{35}$$
$$= \frac{3}{j\omega+2} - \frac{3}{j\omega+3} \tag{36}$$

Recall that $\mathcal{F}[tx(t)] = j\frac{dX(j\omega)}{d\omega}$. Then

$$H_1(j\omega) = \frac{3}{(j\omega+3)^2} = 3j\left(-\frac{j}{(j\omega+3)^2}\right) \tag{37}$$
$$= 3j\frac{d}{d\omega}\left(\frac{1}{j\omega+3}\right) \tag{38}$$
$$= 3j\frac{d\mathcal{F}[e^{-3t}u(t)]}{d\omega} \tag{39}$$

Thus $h_1(t) = 3te^{-3t}u(t)$.

$$H_2(j\omega) = \frac{3}{2+j\omega} + j\omega\frac{1}{2+j\omega} \tag{40}$$

Then

$$h_2(t) = 3e^{-2t}u(t) + \frac{d}{dt}(e^{-2t}u(t)) \tag{41}$$
$$= 3e^{-2t}u(t) - 2e^{-2t}u(t) + \delta(t)e^{-2t} \tag{42}$$

Using the inverse Fourier Transforms from the catalog, we get

$$h(t) = 3(e^{-2t} - e^{-3t})u(t) \tag{43}$$
$$h_1(t) = 3te^{-3t}u(t) \tag{44}$$
$$h_2(t) = 3e^{-2t}u(t) - 2e^{-2t}u(t) + \delta(t)e^{-2t} \tag{45}$$

(b) (9 points) Let $x(t)$ be a real base band signal: $X(j\omega) \neq 0$ for $|\omega| \leq \omega_0$. We process this signal through an LTI system. Let $y(t)$ denote the corresponding output and let $Y(j\omega)$ denote the Fourier transform of $y(t)$. Does $y(t)$ have frequency components different than those of $x(t)$. i.e., is $Y(j\omega) \neq 0$ for some $|\omega| > \omega_0$? What if we process $x(t)$ through a non-LTI system?

- Is $Y(j\omega) \neq 0$ for $|\omega| > \omega_0$?

  Solution:

  Since the system is LTI, $y(t) = h(t) * x(t)$. Then

$$Y(j\omega) = H(j\omega) \cdot X(j\omega) \tag{46}$$

If $X(j\omega) = 0$ for $|\omega| > \omega_0$. Then $Y(j\omega) = 0$ for $|\omega| > \omega_0$. Thus $y(t)$ has the same frequency components as $x(t)$

- What if the system is non-LTI?

  Solution:

  If the system is non-LTI, then $y(t)$ can have different frequency components. For example, if a system shifts the spectrum ($y(t) = x(t)\cos(\omega_c t)$). Then supported regions of $Y(j\omega)$ become $\omega_c \pm \omega$ for $|\omega| \leq \omega_0$

(c) (10 points) Let

$$h_1(t) = \text{sinc}(t/2)\cos(\pi t), \quad h_2(t) = 2\text{sinc}(2t) \tag{47}$$

Input:

$$x(t) = \cos(3\pi t)\cos(4\pi t) \tag{48}$$

Find output of each system. Can we identify an unknown LTI system from one I/O pair? Hint: Recall the input output relationship that we demonstrated if $h(t)$ is real,

$$x(t) = \cos(\omega_0 t) \to y(t) = |H(j\omega_0)|\cos(\omega_0 t + \angle H(j\omega_0)) \tag{49}$$

Solution:

$$\begin{aligned} x(t) &= \cos(3\pi t)\cos(4\pi t) && (50)\\ &= \cos\left(\frac{7\pi t - \pi t}{2}\right)\cos\left(\frac{7\pi t + \pi t}{2}\right) && (51)\\ &= \frac{1}{2}[\cos(7\pi t) + \cos(\pi t)] && (52) \end{aligned}$$

By modulation,

$$\begin{aligned} H_1(j\omega) &= \frac{1}{2}\left[2\text{rect}\left(\frac{\omega - \pi}{\pi}\right) + 2\text{rect}\left(\frac{\omega + \pi}{\pi}\right)\right] && (53)\\ &= \text{rect}\left(\frac{\omega - \pi}{\pi}\right) + \text{rect}\left(\frac{\omega + \pi}{\pi}\right) && (54) \end{aligned}$$

$$H_2(j\omega) = \frac{1}{2} \cdot 2 \cdot \text{rect}(\frac{\omega}{4\pi}) = \text{rect}(\frac{\omega}{4\pi}) \tag{55}$$

Using (49), if we plug in 7π and π (frequencies) into $H_1(j\omega)$ and $H_2(j\omega)$,

$$\begin{aligned} y_1(t) &= \frac{1}{2}|H_1(7\pi)|\cos(7\pi t + \angle H_1(7\pi)) + \frac{1}{2}|H_1(\pi)|\cos(\pi t + \angle H_1(\pi)) && (56)\\ &= \frac{1}{2}\cos(\pi t) && (57) \end{aligned}$$

$$y_2(t) = \frac{1}{2}|H_2(7\pi)|\cos(7\pi t + \angle H_2(7\pi)) + \frac{1}{2}|H_2(\pi)|\cos(\pi t + \angle H_2(\pi)) \tag{58}$$

$$= \frac{1}{2}\cos(\pi t) \tag{59}$$

Since $y_1(t) = y_2(t)$, we can't identify a specific LTI system from its input-output pair because different systems can have the same response.

3. Filters (18 points)

(a) (6 points) Consider ideal low-pass filter $h_{\text{LP},1}(t)$. Construct a system: We are given two choices for α : 1 or -1. Which value should we choose so that the new system is a high-pass filter? Does the new filter have any phase in its frequency response?

Solution:

From the diagram

$$h_{eq} = \alpha\delta(t) + h_{LP,1}(t) \tag{60}$$

$$\implies H_{eq}(j\omega) = \alpha + H_{LP,1}(j\omega) \tag{61}$$

If we want the system to be a high pass filter, we need to cancel out the "rect" part of $H_{LP,1}(j\omega)$. Thus $\alpha = -1$ leads to

$$H_{eq}(j\omega) = \begin{cases} -1, & |\omega| \geq 2\pi \\ 0, & \text{o/w} \end{cases} \tag{62}$$

The phase is π because $H_{eq}(j\omega) = -1$ for all supported regions.

(b) (3 points) Why are ideal filters non-realizable?

Solution:

Since the impulse response is nonzero at $t = 0$, it isn't causal, which makes it non-realizable. Also, since the frequency response is a rect function, the impulse response would require infinite responses in time, which you can't convolve.

(c) (5 points) Design a causal non-ideal LPF with:

$$H_{\text{LP},2}(j\omega) = \frac{k}{\beta + j\omega} \tag{63}$$

Require:

$$|H_{\text{LP},2}(0)| = 1, \quad |H_{\text{LP},2}(j2\pi)| = \frac{1}{\sqrt{2}} \tag{64}$$

Solve for k, β.

Solution:

If $H_{\text{LP},2}(0) = 1$, then $k = \beta$. By the second condition

$$\left|\frac{\beta}{\beta + 2\pi}\right| = \frac{|\beta|}{\sqrt{\beta^2 + (2\pi)^2}} = \frac{1}{\sqrt{2}} \tag{65}$$

Thus

$$2\beta^2 = \beta^2 + 4\pi^2 \tag{66}$$

$$\implies \beta^2 = 4\pi^2 \tag{67}$$

Then $\beta = 2\pi$ as $\beta = -2\pi$ would lead to a noncausal system.

(d) (4 points) Replace ideal LPF in part (a) with $H_{\text{LP},2}(j\omega)$. Does this system act as a high-pass filter?

Solution:

$$H_{eq}(j\omega) = -1 + \frac{2\pi}{2\pi + j\omega} \tag{68}$$

$$= -\frac{2\pi + j\omega}{2\pi + j\omega} + \frac{2\pi}{2\pi + j\omega} \tag{69}$$

$$= -\frac{j\omega}{2\pi + j\omega} \tag{70}$$

$$\implies |H_{eq}(j\omega)| = \frac{|\omega|}{\sqrt{4\pi^2 + \omega^2}} \tag{71}$$

When ω is close to 0, the magnitude of the frequency response will also be close to 0. If ω is really big, then $|H_{eq}(j\omega)| \approx 1$. Thus the system is a high pass filter.

4. Moving Average Filters (10 points)

Impulse response is a boxcar.

(a) (7 points) Find the frequency response $H(j\omega)$.

Solution:

The impulse response is given as $\frac{1}{T}\text{rect}(\frac{t-\frac{T}{2}}{T})$. Then using the catalog, (using modulation)

$$H(j\omega) = e^{-j\omega\frac{T}{2}} \cdot T \cdot \frac{1}{T}\text{sinc}\left(\frac{\omega T}{2\pi}\right) \tag{72}$$

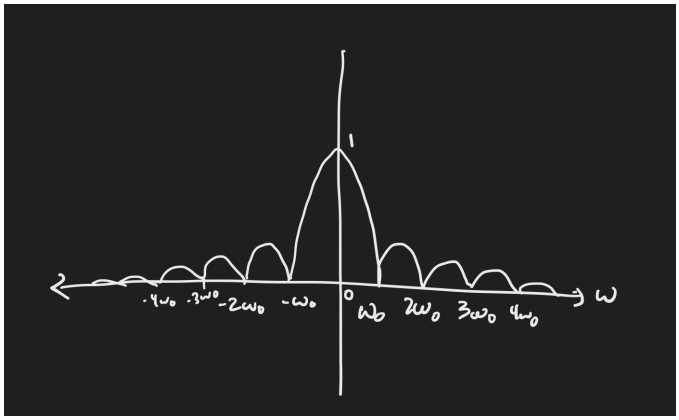
$$= e^{-j\omega\frac{T}{2}}\text{sinc}\left(\frac{\omega T}{2\pi}\right) \tag{73}$$

(b) (3 points) Sketch amplitude response $|H(j\omega)|$. What happens as $\omega \to \infty$?

Solution:

Since complex exponentials have a magnitude of 1, $|H(j\omega)| = \text{sinc}(\frac{\omega T}{2\pi})$. If $\omega_0 = \frac{2\pi}{T}$ and $\omega \to \infty$, $|H(j\omega)| \to 0$.

Amplitude response $|H(j\omega)|$:



5. Modulation and Demodulation (25 points)

(a) (15 points)

Double sideband (DSB) system:

- Modulate $m(t)$ with $\cos(\omega_c t) \to x(t)$
- Pass through ideal bandpass filter $H(j\omega) \to y(t)$
- At receiver, multiply $y(t)$ by $\cos(\omega_c t)$ to get $z(t)$

Sketch:

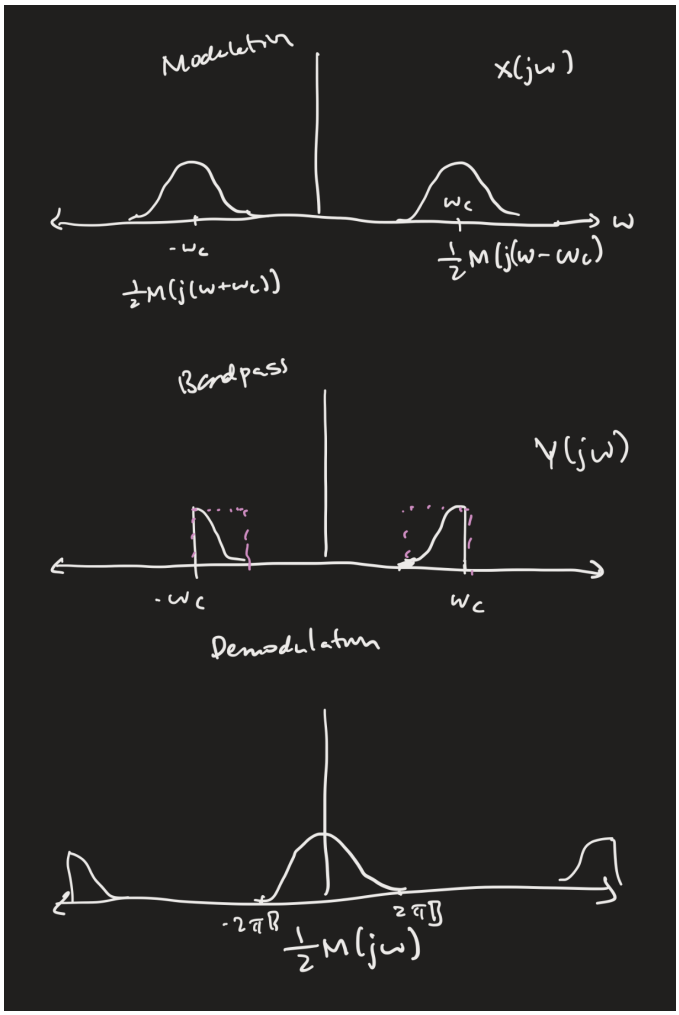
(a) $X(j\omega)$

(b) $Y(j\omega)$

(c) $Z(j\omega)$

Does the system recover $m(t)$?

Solution:



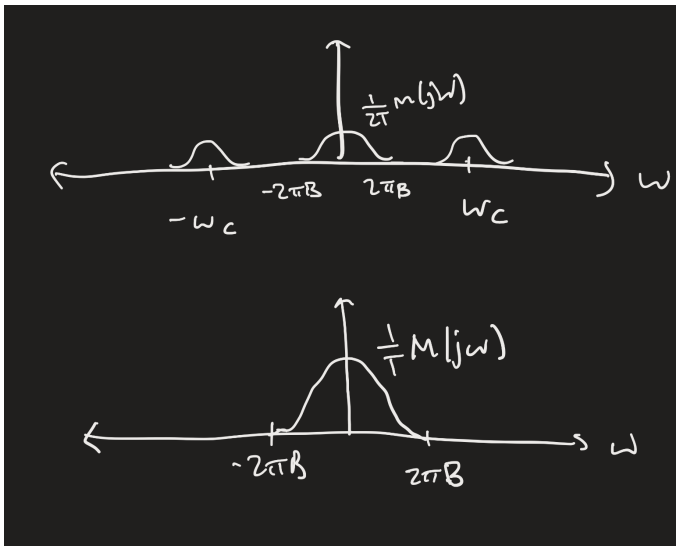
The system does recover $m(t)$. The lowpass filter at the end would remove the ends.

(b) (10 points)

Now replace demodulation with ideal sampler at frequency ω_c.

- Show that this recovers $m(t)$
- Sketch spectrum $Z(j\omega)$ before lowpass filter

Solution:



$Y(j\omega)$ gets repeated every ω_c, which is shown is $Z(j\omega)$ (Top graph). Then after a low pass filter, the system recovers $m(t)$.